

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457893.7419 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.012
Transit depth (Rp/Rs)^2	3.14 +/- 0.43 [%]
Orbital Inclination (inc)	88.39 +/- 0.67
Airmass coefficient 1 (a1)	1.314 +/- 0.014
Airmass coefficient 2 (a2)	-0.199 +/- 0.036
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	None
Optimal Aperture	4.61
Optimal Annulus	12.1
Transit Duration (day)	0.1275 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.177 $\pm$ 0.012 vs 0.1457 (deviation 2.61 σ)
Duration fit vs published	0.1275 d vs 0.12 d (deviation 1.63 σ)
Mid-time O-C	1.5 min
Depth SNR	14.7
Residual scatter	1.07 %

Light curve

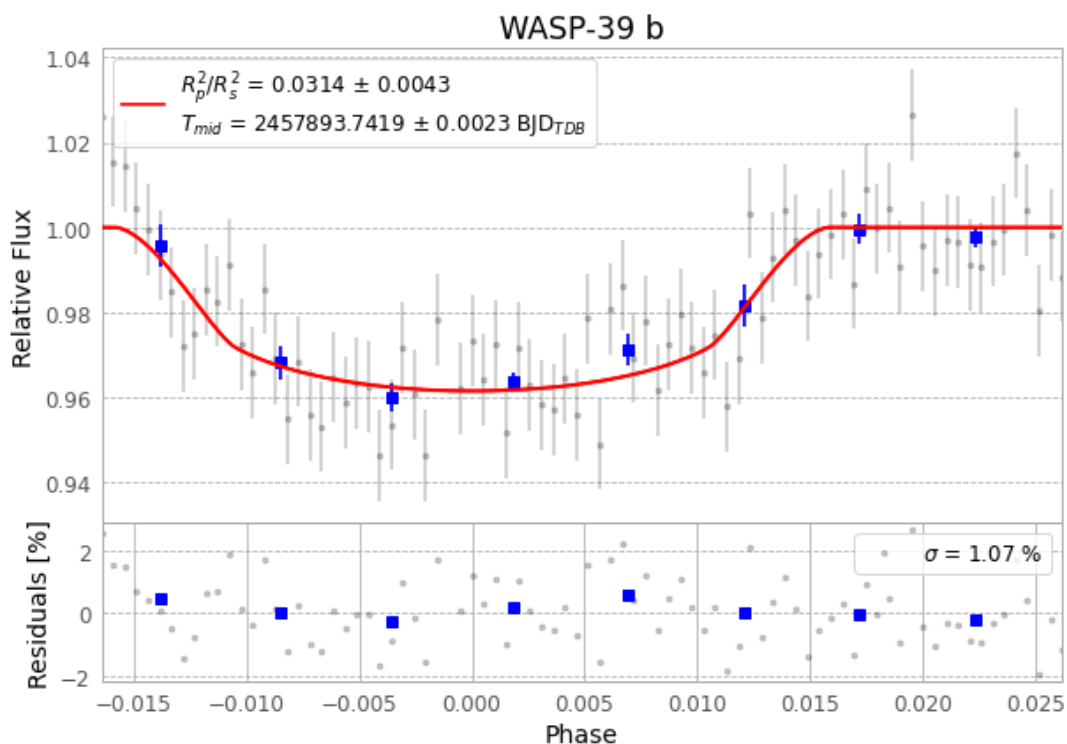


Figure 1 — FinalLightCurve_WASP-39 b_2017-05-19.png (SHA-256 d385106af52326e1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [418, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dd688d49ed380ab2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), WASP-39170520040312.FITS → WASP-39170520081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-170520.log (88d49b81b6c7...), FinalLightCurve_WASP-39 b_2017-05-19.png (d385106af523...), FinalLightCurve_WASP-39 b_2017-05-19.csv (d7e9dbfab82c...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 19:34Z.